

(c) Marks in Chemistry ≥ 40

(d) Total in all 3 subjects ≥ 200 (Or) Total in Maths & Physics ≥ 150

Given the marks in the 3 subjects of n (user input) students, write a program to process the applications to list the eligible candidates.

- Q3. A. What is difference between a class and object? Write a program that demonstrates the following: object creation, accessing object variable and methods. 6
- B. What do you mean by information hiding? Explain how java handles it with a proper example. 6
- C. Explain the following from java perspective 6
1. Stacks and Heaps
 2. Instance variables and Local variables
- Q4. A. What is inheritance tree? Explain its rules of designing with a proper example. 6
- B. Write a program to create a class named Vehicle having protected instance variables regnNumber, speed, color, ownerName and a method showData() to show "This is a vehicle class". Inherit the Vehicle class into subclasses named Bus and Car having individual private instance variables routeNumber in Bus and manufacturerName in Car and both of them having showData() method showing all details of Bus and Car respectively with content of the super class's showData() method. 6
- C. Explain following terminologies with a suitable example: 6
1. Abstract class
 2. Final method
 3. This
- Q5. A. What is interface? Justify the statement with a suitable example: In java via abstract class one can do abstraction, so justify why one need interface to achieve abstraction? 6
- B. Write a Java program that shows the use of all keywords for exception handling. 6
- C. What are threads in java? Explain its life cycle. How threads are created in java? Explain with a suitable example 6
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